

Computational Finance – Final Project

Employee Stock Options

Jelmer Wieringa

May 28, 2026

Employee stock options (ESOs) are a form of compensation that give employees the right to buy company stock at a predetermined strike price. ESOs typically feature a vesting period, during which the option cannot be exercised, followed by an exercise window in which the employee can choose to exercise at any time. ESOs may also have more complex features, such as stochastic strikes or dividends.

Suppose an ESO is granted at $T_0 = 0$, with a vesting period ending at T_1 and final maturity at $T_2 > T_1$. The underlying stock S follows geometric Brownian motion (GBM):

$$\begin{aligned} dS(t) &= S(t) \cdot [r \, dt + \sigma \, dW^{\mathbb{Q}}(t)] \\ &= r \cdot S(t) \, dt + \sigma \cdot S(t) \, dW^{\mathbb{Q}}(t). \end{aligned} \tag{1}$$

1 Backward Induction Routine

Formulate the backward induction (dynamic programming) routine for pricing the ESO, accounting for the vesting period $[T_0, T_1)$ (during which exercise is not allowed) and the exercise window $[T_1, T_2]$ (during which early exercise is possible). Assume the strike is $K = S(T_0)$. Clearly state the value function at each stage.

1.1 Setup & Time Descretization

An Employee Stock Option (ESO) is granted at $T_0 = 0$, has a vesting period $[T_0, T_1)$ during which exercise is forbidden, and an exercise period $[T_1, T_2]$ during which the option holder is allowed to exercise at any time up to maturity T_2 . The strike price is fixed at $K = S(T_0) := S_0$. Under the risk-neutral measure $\mathbb{Q}$, the underlying stock price $S(t)$ follows geometric Brownian motion (GBM) with constant risk-free rate $r \in \mathbb{R}_{>0}$ and constant volatility $\sigma \in \mathbb{R}_{>0}$:

$$dS(t) = r \cdot S(t) \, dt + \sigma \cdot S(t) \, dW^{\mathbb{Q}}(t), \tag{2}$$

where $W^{\mathbb{Q}}(t)$ is standard Brownian motion under $\mathbb{Q}$. The money-market account satisfies the following equation:

$$dM(t) = r \cdot M(t) \, dt, \tag{3}$$

with solution $M(t) = e^{r \cdot t}$ with $M(T_0) = M(0) = 1$. Hence, $M(\tilde{t}_1)/M(\tilde{t}_2) = e^{-r \cdot (\tilde{t}_2 - \tilde{t}_1)}$ is the discount factor for $\tilde{t}_1, \tilde{t}_2 \in [T_0, T_2]$ such that $\tilde{t}_2 \geq \tilde{t}_1$. Moreover, since this option belongs to the class of call options, the payoff function upon exercise is given by:

$$H(S) := \max\{S - K, 0\}. \tag{4}$$

Since the ESO can be exercised continuously in $[T_1, T_2]$, the ESO is an American-style option on that interval. Therefore, to make the estimation procedure computationally tractable, we approximate it by a Bermudan option on a discrete, equidistant time grid of $[T_1, T_2]$. In other words, we discretise the time period $[T_0, T_2]$ into $m \in \mathbb{N}$ equal steps of length $\Delta t = (T_2 - T_0)/m = T_2/m$, yielding the following grid $\mathcal{T}$ with $t_i = i \cdot \Delta t$:

$$\begin{aligned}\mathcal{T} &:= \{t_0 = T_0 = 0, t_1, \dots, t_{m-1}, t_m = T_2\} = \{t_i\}_{i=0}^m \\ &= \{t_i \in \mathcal{T} \mid T_0 \leq t_i < T_1\} \cup \{t_i \in \mathcal{T} \mid T_1 \leq t_i \leq T_2\} \\ &=: \mathcal{T}_{\text{vesting}} \quad \cup \quad \mathcal{T}_E.\end{aligned}\tag{5}$$

We require that $T_1 \in \mathcal{T}$, i.e., the start of the exercise period falls on a grid point.

Furthermore, similar to Longstaff and Schwartz (2001), let $(\Omega, \mathcal{F}, \mathbb{Q})$ be a complete probability space, equipped with the filtration $\{\mathcal{F}_t\}_{t \geq 0}$ generated by the standard Brownian motion $W^\mathbb{Q}(t)$. $\mathbb{Q}$ denotes the risk-neutral measure, under which discounted asset prices are martingales.

1.2 Optimal Decision Strategy

The objective is to determine the price of the ESO at grant ($T_0 = 0$). Since the ESO can only be exercised in $\mathcal{T}_E$, the option's value at T_0 is the highest payoff discounted back to T_0 (Oosterlee and Grzelak, 2019):

$$\begin{aligned}V(T_0, S_0) &= \sup_{\tau \in \mathcal{T}_E} \mathbb{E}^\mathbb{Q} \left[\frac{M(T_0)}{M(\tau)} \cdot H(S(\tau)) \mid \mathcal{F}(T_0) \right] \\ &= \sup_{\tau \in \mathcal{T}_E} \mathbb{E}^\mathbb{Q} \left[e^{-r \cdot \tau} \cdot H(S(\tau)) \mid S(T_0) \right].\end{aligned}\tag{6}$$

τ is the optimal stopping time for the holder to exercise the option. Exercising the option early, i.e., before time T_2 , should maximise the expected payoff at time T_0 . Notice that $\tau \in \mathcal{T}_E$ ensures that exercising in the vesting period is excluded.

Moreover, since S follows GBM, it is a Markov process under $\mathbb{Q}$. Hence, all the information available up to and including time T_0 denoted by the filtration at time T_0 , i.e., $\mathcal{F}(T_0)$, is equal to $S(T_0)$. Analogously, conditioning on $\mathcal{F}(t_i)$ is equivalent to conditioning on $S(t_i)$, for $t_i \in \mathcal{T}$.

1.3 Backward Induction Algorithm

The optimal decision problem, defined in equation (6), can be solved by means of backward dynamic programming (Longstaff and Schwartz, 2001; Oosterlee and Grzelak, 2019). In order to construct the backward induction algorithm, we partition into three phases:

- **Vesting period** ($t_i \in \mathcal{T}_{\text{vesting}}$): During this period, exercising is contractually forbidden. Hence, the holder must continue to hold on to his option. The option's value function $V(t_i, S(t_i))$ is the continuation value denoted by $c(t_i, S(t_i))$.
- **Exercise period** ($t_i \in \mathcal{T}_E \setminus \{T_2\}$): The holder faces the dilemma of whether to exercise or not. Hence, the holder should compare immediate exercise, which yields $H(S(t_i))$, against the expected future payoff, which is based on the continuation of the option. It is optimal to exercise if and only if the immediate payoff $H(S(t_i))$ is larger than (or equal to) the continuation value $c(t_i, S(t_i))$. Note that we assume that a rational holder never exercises when the option is out-of-the-money (OTM), i.e., when $H(S(t_i)) = 0$.
- **Maturity** ($t_i = T_2$): At maturity, the ESO expires and the holder receives $H(S(T_2))$, regardless of whether the option is in-the-money (ITM).

The continuation value of the option at time t_i can be expressed as the conditional expectation of the discounted option value at a later time t_{i+1} :

$$\begin{aligned} c(t_i, S(t_i)) &:= \mathbb{E}^{\mathbb{Q}} \left[\frac{M(t_i)}{M(t_{i+1})} \cdot V(t_{i+1}, S(t_{i+1})) \mid \mathcal{F}(t_i) \right] \\ &= e^{-r \cdot \Delta t} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(t_{i+1}, S(t_{i+1})) \mid S(t_i) \right]. \end{aligned} \quad (7)$$

The continuation value represents the expected gain from not exercising at t_i

Hence, the backward induction algorithm, from $t_m = T_2$ to $T_0 = 0$, is given by:

$$V(t_i, S(t_i)) = \begin{cases} c(t_i, S(t_i)), & t_i \in \mathcal{T}_{\text{vesting}}; \\ \max \{ H(S(t_i)), c(t_i, S(t_i)) \}, & t_i \in \mathcal{T}_E \setminus \{T_2\}; \\ H(S(T_2)), & t_i = T_2. \end{cases} \quad (8)$$

1.4 ESO Price at Grant T_0

Iterating the algorithm in equation (8) from $t_m = T_2$ to $T_0 = 0$ yields the value of the option at T_0 , i.e., $V(T_0, S_0)$. Since exercising is forbidden in the vesting period and $T_0 \in \mathcal{T}_{\text{vesting}}$, $V(T_0, S_0)$ can be rewritten using the tower rule:

$$\begin{aligned} V(T_0, S_0) &= c(T_0, S_0) = e^{-r \cdot \Delta t} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(t_1, S(t_1)) \mid S(T_0) \right] \\ &= e^{-r \cdot \Delta t} \cdot \mathbb{E}^{\mathbb{Q}} \left[e^{-r \cdot \Delta t} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(t_2, S(t_2)) \mid S(t_1) \right] \mid S(T_0) \right] \\ &= e^{-r \cdot 2 \cdot \Delta t} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(t_2, S(t_2)) \mid S(T_0) \right] \\ &= \dots \\ &= e^{-r \cdot (T_1 - T_0)} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(T_1, S(T_1)) \mid S(T_0) \right] = e^{-r \cdot T_1} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(T_1, S(T_1)) \mid S(T_0) \right]. \end{aligned} \quad (9)$$

2 Least Squares Monte Carlo (LSM) Algorithm

Implement a regression Monte Carlo algorithm to estimate the price of the ESO at grant (T_0) with the above features. Provide a numerical example using chosen parameters (S_0, r, σ, T_1, T_2).

First, we illustrate the mechanics of the LSM algorithm in similar fashion to Longstaff and Schwartz (2001). The corresponding Python code is presented in `CF_Project_2_NumericalExample.py`. Second, we interpret the results of the generalized code in `CF_Project_2.py`. These results include convergence plots, bases comparisons and sensitivity plots.

2.1 Simple Numerical Example

Let $S_0 = K := 1$, $r = 0.06$, and $\sigma = 0.06$. We consider a time grid of four points:

$$\mathcal{T} := \{T_0 := 0, T_1 := 1, T_{3/2} := 2, T_2 := 3\}. \quad (10)$$

First, we generate stock price paths. Since S , defined in equation (29), follows GBM, the exact solution of S at t_i is given by (Oosterlee and Grzelak, 2019):

$$\begin{aligned} S(t_{i+1}) &= S(t_i) \cdot \exp \left\{ \left(r - \frac{1}{2}\sigma^2 \right) \cdot [(t_{i+1} - (t_i)] + \sigma \cdot [W^{\mathbb{Q}}((t_{i+1}) - W^{\mathbb{Q}}((t_i))] \right\} \\ &= S_0 \cdot \exp \left\{ \left(r - \frac{1}{2}\sigma^2 \right) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z_i \right\}, \quad Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \end{aligned} \quad (11)$$

where $\Delta t = T_2/m$ and $W^{\mathbb{Q}}((t_{i+1}) - W^{\mathbb{Q}}((t_i) \sim \mathcal{N}(0, \Delta t)$. Applying equation (11) repeatedly from T_0 to T_2 across $N = 5$ independent paths yields the matrix of simulated stock prices presented in Table 1. We used `np.random.seed(23)`. The green and red numbers denote paths that are ITM and OTM at time t_i , respectively. Note that a call option is ITM at t_i , if $K < S(t_i)$, or equivalently: $H(S(t_i)) = \max\{S(t_i) - 1, 0\} > 0$. Moreover, we denote the paths in the rows at time t_i by $S^{(j)}(t_i)$, for $j \in \{1, \dots, N\}$. Therefore, the matrix of stock prices can also be represented as $\{S^{(j)}(t_i)\}_{i=0, j=1}^{m, N} = \{\mathbf{S}(t_i)\}_{i=0}^m$.

Table 1: $N = 5$ simulated stock price paths.

STOCK PRICE PATHS				
Path	$t = T_0 = 0$	$t = T_1 = 1$	$t = T_2 = 2$	$t = T_3 = 3$
1	1	1.19	1.24	1.11
2	1	1.26	1.51	1.27
3	1	0.97	0.8	0.64
4	1	1.48	1.44	1.71
5	1	1.11	1.17	0.99

Our objective is to determine the stopping rule that maximises the value of the option at each point along the paths. We use the dynamic programming routine given in equation (8) to determine the optimal strategy. In particular, we use the backward induction algorithm to fill in a cash flow matrix whose columns correspond to $\mathcal{T} \setminus \{T_0\}$. Next, we discount each column of

the cash flow matrix back to time T_0 to obtain the discounted cash flows at T_0 according to the optimal strategy for $N = 5$ paths. Finally, to obtain the estimated ESO price at grant, i.e., $\hat{V}_0$, we take the sample average of the discounted cash flows at T_0 .

Second, we determine the optimal decision at maturity $T_2 = 3$. Since there is no choice at maturity, the optimal decision is simply given by the payoff at maturity, i.e., $H(S^{(j)}(T_2))$. This yields the last column of the cash flow matrix which corresponds to time $T_2 = 3$. Note that this is consistent with the backward induction recursion defined in equation (8).

Table 2: Cash flow matrix at $t = T_2 = 3$, with $N = 5$ paths.

CASH FLOW MATRIX AT $t = T_2 = 3$			
Path	$t = T_1 = 1$	$t = T_{3/2} = 2$	$t = T_2 = 3$
1			0.11
2			0.27
3			0
4			0.71
5			0

Third, we determine the optimal decision at time $T_{3/2} = 2$. If the call option is ITM at time $T_{3/2} = 2$, then the holder either exercises immediately or continues the option's life until time $T_2 = 3$. Hence, we need to compute $\max\{H(S^{(j)}(T_{3/2})), c(T_{3/2}, S^{(j)}(T_{3/2}))\}$, for all $j \in \{1, \dots, N\}$. The payoff $H(S^{(j)}(T_{3/2})) = \max\{S^{(j)}(T_{3/2}) - 1, 0\}$ is known. However, the continuation value c , defined in equation (7), is unknown. The LSM algorithm estimates this conditional expectation by ordinary least squares (OLS). Below is a blue box that presents an overview of classical linear regression with OLS.

Let us define $\mathbf{Y}_{t=2}$ as the discounted cash flows received at time $t = T_2 = 3$, if the option is not exercised at $t = T_{3/2} = 2$. Additionally, define $\mathbf{S}_{t=2} = \mathbf{X}_{t=2}$ as the stock prices at $t = T_{3/2} = 2$. Table 3 shows the values for $\mathbf{Y}_{t=2}$ and $\mathbf{S}_{t=2}$. The LSM algorithm only uses ITM paths, so we need to filter out all the OTM paths. Since $S_{t=2}^{(3)}$ is the only element that is below the strike price $K = 1$, hence it is coloured red, we discard the third path since it is OTM. The elements of $\mathbf{Y}_{t=2}$ that correspond to ITM paths are determined by discounting the cashflow at $t = T_2 = 3$ to $t = T_{3/2} = 2$. Next, we estimate the continuation value by OLS with $\{1, S, S^2\}$ as basis:

$$\begin{aligned}
c(T_{3/2}, S^{(j)}(T_{3/2})) &= f(X_j) \approx \hat{f}(X_j) = \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot S_{t=2}^{(j)} + \hat{\beta}_3 \cdot (S_{t=2}^{(j)})^2 \\
&\approx -25.8 + 38.0 \cdot S_{t=2}^{(j)} - 13.7 \cdot (S_{t=2}^{(j)})^2.
\end{aligned} \tag{12}$$

This regression equation enables us to estimate the continuation value. Note that the estimated parameters are unreliable and for illustrative purposes only, because $N = 5$ and $m = 3$ are very small parameter values for convergence. Furthermore, to conclude which path should exercise at time $t = T_{3/2} = 2$ or continue to hold the option until $t = T_2 = 3$, we use the backward induction algorithm. Table 4 suggests that the second and fifth path should exercise at $t = T_{3/2} = 2$. This enables us to fill in the column in the cash flow matrix corresponding to $t = T_{3/2} = 2$, which is denoted in Table 5.

Classical Linear Regression — Overview

Suppose you have n observations of both the dependent variable (response variable) Y_i and independent variable (regressor) X_i . You assume that their relationship is given by:

$$Y_i = f(X_i) + \epsilon_i, \quad i \in \{1, \dots, n\} \quad \& \quad \mathbb{E}[\epsilon_i | X_i] = 0, \quad (13)$$

where $f(X_i) = \mathbb{E}[Y_i | X_i]$. Hence, the objective is to estimate the unknown function f . Fundamentally, regression is about estimating a conditional expectation. However, you cannot estimate an arbitrary function f from a finite sample without assuming some structure. The classical assumption is that f is a linear function in a set of $M \in \mathbb{N}$ known basis functions $\{\psi_1(X_i), \dots, \psi_M(X_i)\}$.

If $\{\psi_k(X_i)\}_{k=1}^{\infty}$ forms a complete orthonormal basis for $L^2(\mu)$, for some measure μ , then every function $f \in L^2(\mu)$ can be written as an infinite combination of basis functions:

$$f(X_i) = \mathbb{E}[Y_i | X_i] = \sum_{k=1}^{\infty} \beta_k \cdot \psi_k(X_i), \quad \beta_k = \int f(x) \cdot \psi_k(x) \, d\mu(x). \quad (14)$$

Hence, $f(X_i)$ can be estimated by truncating the summation:

$$f(X_i) = \mathbb{E}[Y_i | X_i] \approx \sum_{k=1}^M \beta_k \cdot \psi_k(X_i) =: \hat{f}(X_i). \quad (15)$$

So, the linear model can be written as:

$$\mathbf{Y} = \mathbf{\Psi} \cdot \boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0 | \sigma^2 I), \quad (16)$$

where $\mathbf{\Psi} \in \mathbb{R}^{n \times M}$ is the design matrix defined by $\Psi_{ik} := \psi_k(X_i)$.

The basis functions ψ_k are chosen, but the coefficients $\boldsymbol{\beta} := (\beta_1, \dots, \beta_M)^T$ are unknown and must be estimated from the data. The best linear predictor of $\mathbf{Y} := (Y_1, \dots, Y_n)^T = (f(X_1), \dots, f(X_n))^T =: f(\mathbf{X})$, in terms of minimising the mean squared error, is the OLS estimator:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^M} \|\mathbf{\Psi} \boldsymbol{\beta} - \mathbf{Y}\|^2 = (\mathbf{\Psi}^T \mathbf{\Psi})^{-1} \mathbf{\Psi}^T \mathbf{Y}. \quad (17)$$

Thus, $\mathbb{E}[\mathbf{Y} | \mathbf{X}]$ is (OLS) estimated by $\hat{\mathbf{Y}} = \hat{f}(\mathbf{X}) = \mathbf{\Psi} \cdot \hat{\boldsymbol{\beta}}$.

Table 3: Regression data at $t = T_{3/2} = 2$.

REGRESSION DATA AT $t = 2$		
Path	$\mathbf{Y}_{t=2}$	$\mathbf{S}_{t=2}$
1	$e^{-0.06} \cdot 0.11$	1.24
2	$e^{-0.06} \cdot 0.27$	1.51
3	0	0.8
4	$e^{-0.06} \cdot 0.71$	1.44
5	$e^{-0.06} \cdot 0$	1.17

Table 4: Optimal early exercise decision at $t = T_{3/2} = 2$.

OPTIMAL EARLY EXERCISE DECISION AT $t = 2$			
Path	Exercise		Continuation
1	0.24	<	≈ 0.258
2	0.51	>	≈ 0.345
3	—		—
4	0.44	<	≈ 0.514
5	0.17	>	≈ -0.09

Table 5: Cash flow matrix at $t = T_{3/2} = 2$, with $N = 5$ paths.

CASH FLOW MATRIX AT $t = T_{3/2} = 2$			
Path	$t = T_1 = 1$	$t = T_{3/2} = 2$	$t = T_2 = 3$
1		0	0.11
2		0.51	0.27 0
3		0	0
4		0	0.71
5		0.17	0

Fourth, similar to above, we determine the optimal decision at time $T_1 = 1$. The main difference is that $\mathbf{Y}_{t=1}$ is defined as the discounted cash flows received from the optimal continuation strategy, if the option is not exercised at $t = T_1 = 1$. The optimal continuation strategy is determined in the cash flow matrix given in Table 5. Hence, $\mathbf{Y}_{t=1}$ is computed by discounting the columns of Table 5 to time $t = T_1 = 1$:

$$Y_{t=1}^{(j)} = e^{-r \cdot \Delta t} \cdot CF[j, t = 2] + e^{-r \cdot 2 \cdot \Delta t} \cdot CF[j, t = 3], \quad (18)$$

where $CF[j, t]$ denotes the element in the j^{th} row of the column corresponding to t of the cash flow matrix. Equivalently, $CF[j, t]$ denotes that path j receives the undiscounted payoff $H(S(t))$ at time t .

The stock prices at $t = T_1 = 1$ are contained in $\mathbf{S}_{t=1} = \mathbf{X}_{t=1}$. Table 6 depicts the regression data at $t = T_1 = 1$. The third path is omitted because it is OTM. Applying OLS regression yields the following estimation of the continuation value:

$$\begin{aligned} c(T_1, S^{(j)}(T_1)) &= \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot S_{t=1}^{(j)} + \hat{\beta}_3 \cdot (S_{t=1}^{(j)})^2 \\ &\approx -3.98 + 5.34 \cdot S_{t=1}^{(j)} - 1.50 \cdot (S_{t=1}^{(j)})^2. \end{aligned} \quad (19)$$

This allows us to determine for which path it is optimal to exercise at time $t = T_1 = 1$. Table 7 shows that it is only optimal for the fifth path to exercise at $t = T_1 = 1$. Table 8 presents the final cash flow matrix at $t = T_1 = 1$.

Table 6: Regression data at $t = T_1 = 1$.

REGRESSION DATA AT $t = 1$		
Path	$\mathbf{Y}_{t=1}$	$\mathbf{S}_{t=1}$
1	$e^{-0.06} \cdot 0 + e^{-0.12} \cdot 0.11$	1.19
2	$e^{-0.06} \cdot 0.51 + e^{-0.12} \cdot 0$	1.26
3	$0 + 0$	0.97
4	$e^{-0.06} \cdot 0 + e^{-0.12} \cdot 0.71$	1.48
5	$e^{-0.06} \cdot 0.17 + e^{-0.12} \cdot 0$	1.11

Table 7: Optimal early exercise decision at $t = T_1 = 1$.

OPTIMAL EARLY EXERCISE DECISION AT $t = 1$			
Path	Exercise		Continuation
1	0.19	<	≈ 0.25
2	0.26	<	≈ 0.37
3	—		—
4	0.48	<	≈ 0.64
5	0.11	>	≈ 0.10

Table 8: Cash flow matrix at $t = T_1 = 1$, with $N = 5$ paths.

CASH FLOW MATRIX AT $t = T_1 = 1$			
Path	$t = T_1 = 1$	$t = T_{3/2} = 2$	$t = T_2 = 3$
1	0	0	0.11
2	0	0.51	0.27 0
3	0	0	0
4	0	0	0.71
5	0.11	0.17 0	0

Fifth, we estimate the ESO price at grant. The exact ESO price at grant, $V(T_0, S(T_0))$, is defined in equation (9):

$$V(T_0, S_0) = c(T_0, S_0) = e^{-r \cdot T_1} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(T_1, S(T_1)) \mid S(T_0) \right].$$

We estimate this by taking the sample mean of the discounted columns of the cash flow matrix:

$$\hat{V}(T_0, S_0) := \frac{1}{N} \cdot \sum_{j=1}^N \sum_{i=1}^m e^{-r \cdot t_i} \cdot CF[j, i]. \quad (20)$$

This yields an estimated ESO price at grand of approximately $\hat{V}(T_0, S_0) \approx 0.248$.

2.2 Generalised Example – LSM Algorithm

The code in `CF_Project_2.py` generalises the code in `CF_Project_2_NumericalExample.py` by transforming everything into functions, but the structure and logic remain equivalent to the previous subsection. The parameters are $S_0 = 1$, $K = S_0 = 1$, $r = 0.06$, $\sigma = 0.2$, $T_0 = 0$, $T_1 = 3$, $T_2 = 10$, $m = 120$ ($\Delta t = 1/12$, i.e., monthly steps), and $N = 50,000$ paths.

Figure 6, in the Appendix, depicts $N = 23$ simulated GBM paths of $S(t)$ on $[T_0, T_2] = [0, 10]$, generated via equation (11). The dashed vertical line marks the end of the vesting period $T_1 = 3$, and the dashed horizontal line marks the strike $K = 1$. Paths above (below) K at a given $t_i \in \mathcal{T}_E$ are ITM (OTM). In the LSM algorithm, only ITM paths are considered for the regression step.

Figure 1 reports the estimated ESO price $\hat{V}_0$, with $N \in \{1000, 5000, \dots, 10^5\}$, for six families of basis functions: monomials, Legendre polynomials, unstandardised Laguerre polynomials, standardised Laguerre polynomials, Chebyshev polynomials, and trigonometric functions. Both the unstandardised and standardised Laguerre polynomials are general Laguerre polynomials multiplied by $e^{-S/2}$, similar to Longstaff and Schwartz (2001). Standardising refers to dividing stock price paths by their sample mean. Additionally, each basis functions is used with a specific number of terms, i.e., $n_{Terms} \in \{1, 3, 5\}$. The dashed red line marks the benchmark of monomial basis with three terms for comparison. Furthermore, the filled-in regions next to the coloured lines denote the confidence intervals of $\hat{V}_0$. By the central limit theorem:

$$\frac{\hat{V}_0 - V_0}{\hat{\sigma}/\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (21)$$

where $\hat{\sigma}$ denotes the unbiased sample standard deviation. Hence, the 95% confidence interval of $\hat{V}_0$ is given by:

$$\left[\hat{V}_0 - \phi^{-1}(0.975) \cdot \frac{\hat{\sigma}}{\sqrt{N}}, \quad \hat{V}_0 + \phi^{-1}(0.975) \cdot \frac{\hat{\sigma}}{\sqrt{N}} \right], \quad (22)$$

where ϕ denotes the standard normal cumulative distribution function.

Several conclusions emerge. First, only for monomial, Legendre, standardised Laguerre and Chebyshev bases, $\hat{V}_0$ converges as N increases, for sufficiently large n_{Terms} . Second, for $M = 1$, $\hat{V}_0$ is systematically underestimated across all families. This confirms that at least a linear term in S is needed. Third, $n_{Terms} = 3$ and $n_{Terms} = 5$ produce similar estimates for monomials, Legendre, and Chebyshev bases. This indicates that three terms are sufficient for these well-conditioned bases. Fourth, the unstandardised Laguerre basis shows a persistent gap relative to the monomial benchmark even at $n_{Terms} = 5$, whereas standardising it closes this gap significantly. Summarising, the monomial basis with $n_{Terms} = 3$ provides a reliable choice, consistent with Longstaff and Schwartz (2001).

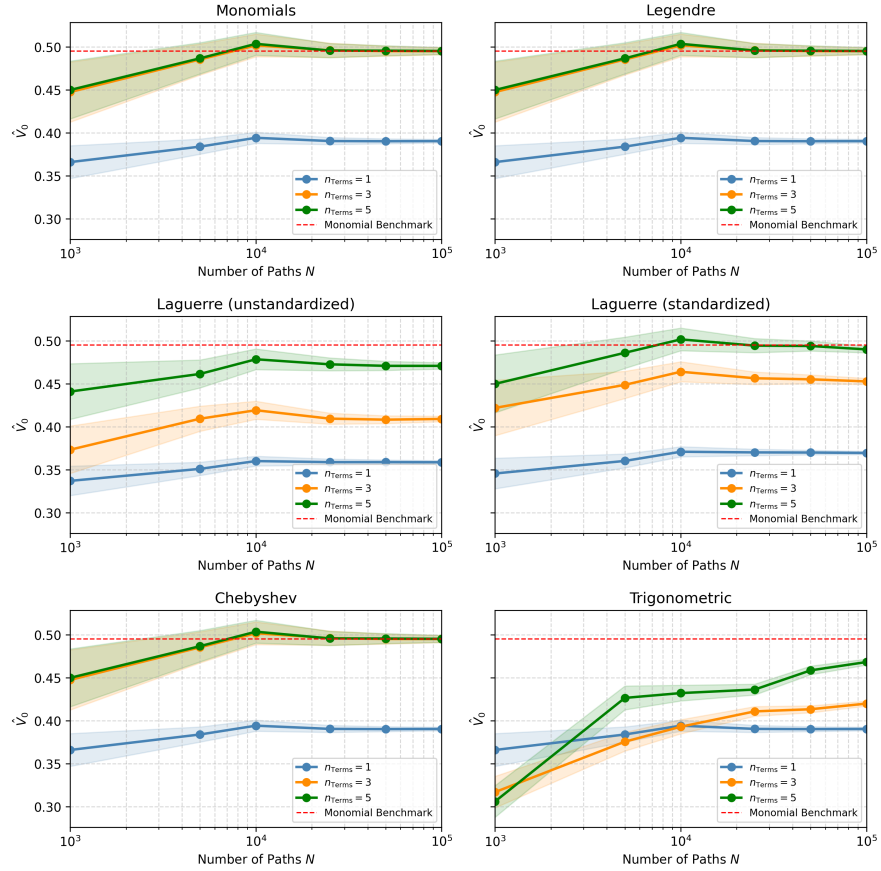


Figure 1: Convergence of $\hat{V}_0$ as a function of N for six basis function families with $n_{Terms} \in \{1, 3, 5\}$. The dashed red line marks the monomial benchmark ($M = 3$).

Figure 7, in the Appendix, plots $\hat{V}_0$ as a function of volatility $\sigma \in [0.05, 1.0]$, holding all other parameters fixed. The ESO price is increases in σ on average. This is because higher volatility increases the probability of large upside moves, increasing the option's expected payoff.

Figure 8, in the Appendix, plots $\hat{V}_0$ as a function of the left exercise boundary $T_1 \in [0, T_2)$, holding $T_2 = 10$ and all other parameters fixed. $\hat{V}_0$ is effectively constant in T_1 : the total variation across the entire range $T_1 \in [0, T_2)$ is of order 10^{-4} . This is because early exercise is never optimal for a call option on a non-dividend paying stock. Since holding the option always preserves time value that would be destroyed upon exercise, the continuation value strictly dominates the exercise value at every $t_i \in \mathcal{T}_E \setminus \{T_2\}$.

3 Stochastic Strike K

Consider a modification in which the strike is set at $K = S(T_{1/2})$, where $T_{1/2} \in [T_0, T_1]$ (specified a priori within the vesting period). At grant, the strike is unknown and will be fixed at $T_{1/2}$. Modify your pricing routine to accommodate this feature, and discuss how the ESO price changes as a function of $T_{1/2}$.

In the previous section, the strike is the fixed constant $K = S_0 = S(T_0)$. Here, the strike price is instead the stochastic stock price at a pre-specified date $T_{1/2} \in [T_0, T_1]$ within the vesting period:

$$K^{(j)} := S^{(j)}(T_{1/2}), \quad j = 1, \dots, N. \quad (23)$$

Note that $T_{1/2}$ is known at T_0 because it is contractually determined. The strike price is known at $T_{1/2}$ as $K = S(T_{1/2})$. Crucially, $K^{(j)}$ varies across paths, but is constant along paths. Hence, we collect all stochastic strikes in a vector $\mathbf{K} = (K^{(1)}, \dots, K^{(N)})^T \in \mathbb{R}^N$.

The payoff function is path-dependent and becomes:

$$H^{(j)}(S(t_i)) := \max \left\{ S^{(j)}(t_i) - K^{(j)}, 0 \right\}. \quad (24)$$

Similar to the valuation function $V(t_i, S^{(j)}(t_i, K^{(j)}))$, the continuation value depends on two state variables:

$$c(t_i, S^{(j)}(t_i), K^{(j)}) = e^{-r \cdot \Delta t} \cdot \mathbb{E}^{\mathbb{Q}} \left[V(t_{i+1}, S^{(j)}(t_{i+1}, K^{(j)})) \mid S^{(j)}(t_i), K^{(j)} \right]. \quad (25)$$

With a fixed strike, the continuation value is a function of $S^{(j)}(t_i)$ alone. With a stochastic strike, the moneyness ratio $M^{(n)}(t_i) := S^{(n)}(t_i)/K^{(n)}$ is the natural state variable, since it captures the relative position of the stock price with respect to the path-dependent strike. And it has the advantage of transforming two regressors into one regressor. Two implementations are considered:

- **Regress on S :** Use $S^{(j)}(t_i)$ as the regressor, keeping the monomial basis with three terms $\{1, S, S^2\}$.
- **Regress on $M = S/K$:** Normalise the stock price by the path-dependent strike and use $M^{(n)}(t_i) = S^{(n)}(t_i)/K^{(n)}$ as the regressor with basis $\{1, M, M^2\}$.

The parameters are identical to the previous section, with $T_{1/2} = T_1/2 = 1.5$ as the default strike date. Table 9 shows the estimated ESO prices for the fixed-strike baseline and both stochastic strike implementations.

Table 9: LSM pricing results: fixed strike vs. stochastic strike ($T_{1/2} = T_1/2 = 1.5$).

Case	$\hat{V}_0$	95% CI
Fixed strike $K = S(T_0)$	0.4952	[0.4893, 0.5010]
Stochastic strike, regress on S	0.3980	[0.3935, 0.4025]
Stochastic strike, regress on M	0.4098	[0.4051, 0.4146]

The estimated ESO price at grant for the stochastic strike price is substantially lower than for the fixed strike case. This is because, under the risk-neutral measure $\mathbb{Q}$, the stock drifts upward at rate r . Hence:

$$\mathbb{E}^{\mathbb{Q}} [K^{(j)}] = \mathbb{E}^{\mathbb{Q}} [S(T_{1/2})] = S_0 \cdot e^{r \cdot T_{1/2}} = e^{0.06 \cdot 1.5} \approx 1.094 > S_0 = 1. \quad (26)$$

On average, the stochastic strike is set above the fixed strike $K = S_0 = 1$. Since the ESO is a call option, a higher strike directly reduces the expected payoff at every exercise date, lowering the option's value.

Furthermore, regarding the two regression variants, the moneyness regressor, i.e., $M = S/K$ with $\hat{V}_0 \approx 0.410$, produces a higher estimate than the simple stock price regressor, i.e., S with $\hat{V}_0 \approx 0.398$. The confidence intervals are non-overlapping. This is explained by the fact that the continuation value depends on how far ITM the option is relative to its own strike, not on the absolute level of S . Therefore, the basis $\{1, M, M^2\}$ approximates the continuation value function more accurately, yielding a less downward-biased price estimate.

3.1 Sensitivity to $T_{1/2}$

Figure 2 plots $\hat{V}_0$ as a function of $T_{1/2} \in [T_0, T_1)$ for both regression variants.

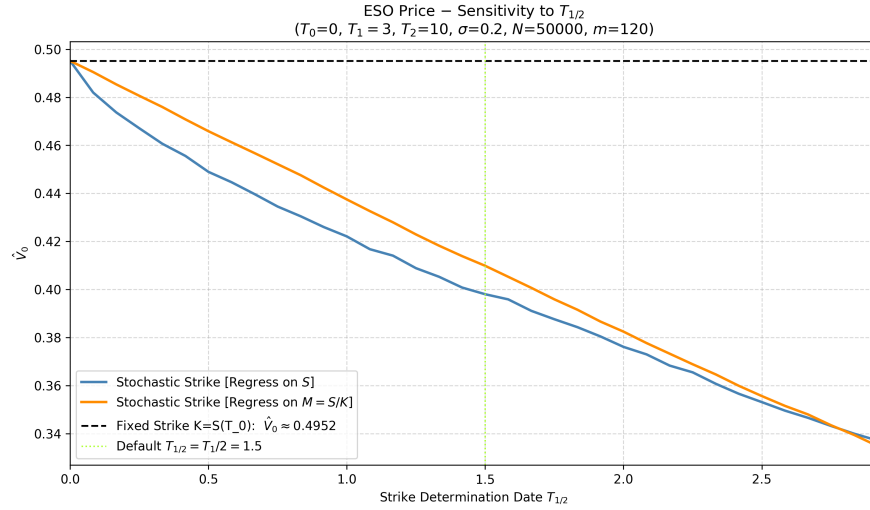


Figure 2: Estimated ESO price $\hat{V}_0$ as a function of the strike determination date $T_{1/2} \in [T_0, T_1)$. The dashed black line marks the fixed-strike baseline $K = S(T_0)$.

The price is decreasing in $T_{1/2}$. First, as $T_{1/2}$ increases, the strike $K^{(n)} = S^{(n)}(T_{1/2})$ is set later, incorporating more of the stock's random fluctuation. Since GBM has increasing variance, a later $T_{1/2}$ produces a more dispersed distribution of strikes. Note that a high strike severely reduces value. Second, as $T_{1/2} \rightarrow T_0$, the stochastic strike converges to the fixed strike $K = S_0$. Furthermore, both regression variants agree closely.

3.2 (Empirical) Distribution of Strike K

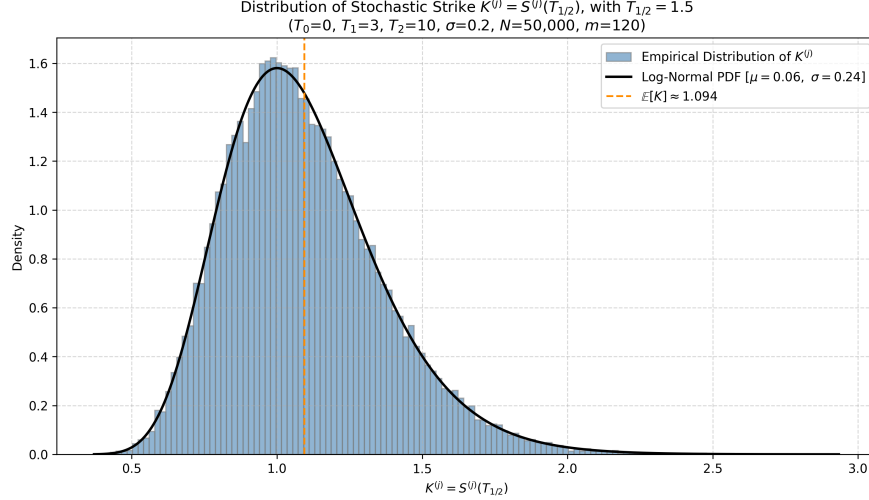


Figure 3: Empirical distribution of the path-specific strike $K^{(j)} = S^{(j)}(T_{1/2})$ for $T_{1/2} = 1.5$, overlaid with the theoretical log-normal density. The dashed orange line marks $\mathbb{E}^{\mathbb{Q}}[K]$.

Figure 3 displays the empirical distribution of the path-specific strike $K^{(j)} = S^{(j)}(T_{1/2})$ across $N = 50000$ paths, with $T_{1/2} = 1.5$. For this histogram, we use the Freedman-Diaconis rule for the optimal bin width choice:

$$h = 2 \cdot IQR(\mathbf{K}) \cdot N^{-1/3}, \quad (27)$$

where h denotes the bin width and IQR denotes the interquartile range.

Since $S(T_{1/2})$ follows GBM, it is log-normally distributed under $\mathbb{Q}$:

$$\begin{aligned} K^{(j)} = S^{(j)}(T_{1/2}) &\sim \text{LogNormal}(\log S_0 + (r - \frac{1}{2}\sigma^2)T_{1/2}, \sigma^2 T_{1/2}) \\ &\stackrel{d}{=} \text{LogNormal}(\mu_K, \sigma_K). \end{aligned} \quad (28)$$

The empirical histogram closely follows the theoretical log-normal density, confirming the simulation is correct.

Two features of this distribution are directly relevant for pricing. First, the mean strike $\mathbb{E}^{\mathbb{Q}}[K] \approx 1.094$ exceeds the fixed strike $K = S_0 = 1$, which is the primary reason that the stochastic strike ESO is cheaper than the fixed strike baseline. Second, the distribution is right-skewed: there is a non-negligible probability mass on relatively large values of $K^{(j)}$. Large values $K^{(j)}$ correspond to paths on which the stock has severely increased by $T_{1/2}$, in which case the option is deep OTM at the start of the exercise period.

4 Dividend Payment d

Suppose the stock pays an annualised continuous dividend d , so that

$$dS(t) = [r - d] \cdot S(t) dt + \sigma \cdot S(t) dW^{\mathbb{Q}}(t), \quad (29)$$

Repeat the pricing experiment for the original ESO ($K = S_0$), and discuss how dividends affect the optimal exercise strategy and ESO value.

With a continuous dividend $d \in \mathbb{R}_{>0}$, the risk-neutral drift of S decreases from r to $r - d$. The exact GBM simulation scheme is the only component that changes:

$$S(t_{i+1}) = S(t_i) \exp \left\{ \left(r - d - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_i \right\}, \quad Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1). \quad (30)$$

4.1 Effect on ESO Price at Grant

Figure 4 plots $\hat{V}_0$ as a function of $d \in [0, 0.06]$, holding all other parameters fixed. The ESO price is strongly decreasing in d , which is driven by two effects. First, a higher dividend yield d reduces the risk-neutral drift of S , which lowers the expected stock price at every future date. Since the ESO is a call option, a lower expected stock price directly reduces the expected payoff. Second, dividends destroy the classical no-early-exercise result. For a non-dividend paying stock, early exercising is never optimal. However, with dividends, the holder loses dividend income by continuing to hold the option rather than exercising and holding the stock. Once d is sufficiently large, the dividend income lost by waiting exceeds the option's remaining time value, and early exercising becomes optimal.

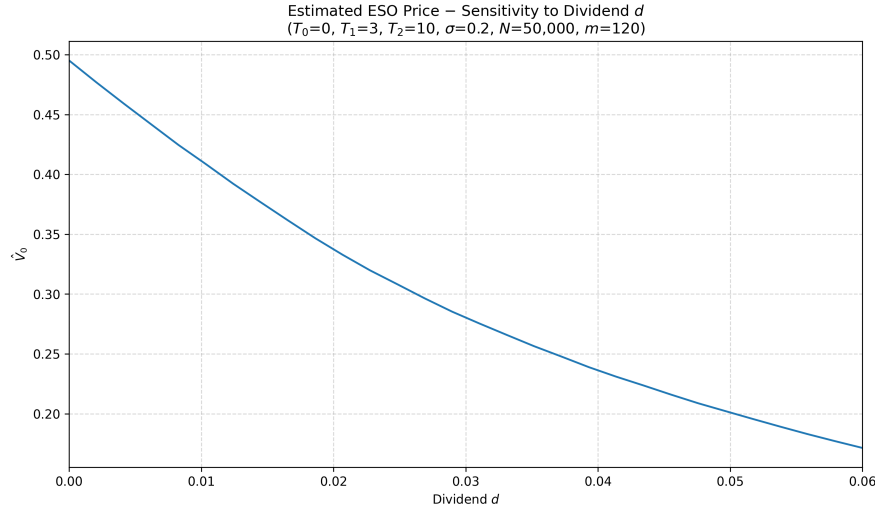


Figure 4: Estimated ESO price $\hat{V}_0$ as a function of dividend $d \in [0, 0.06]$, with all other parameters fixed.

4.2 Sensitivity of Risk-Free Rate r & Dividend d

Figure 5 plots $\hat{V}_0$ over a grid of $(r, d) \in [0.01, 0.1] \times [0, 0.1]$. The surface reveals that the ESO price is increasing in r and decreasing in d . The positive effect of r reflects the fact that a

higher risk-neutral drift $r - d$ raises expected future stock prices. The negative effect of d is unambiguous: it simultaneously reduces the risk-neutral drift and makes early exercise optimal, both of which decrease the ESO price.

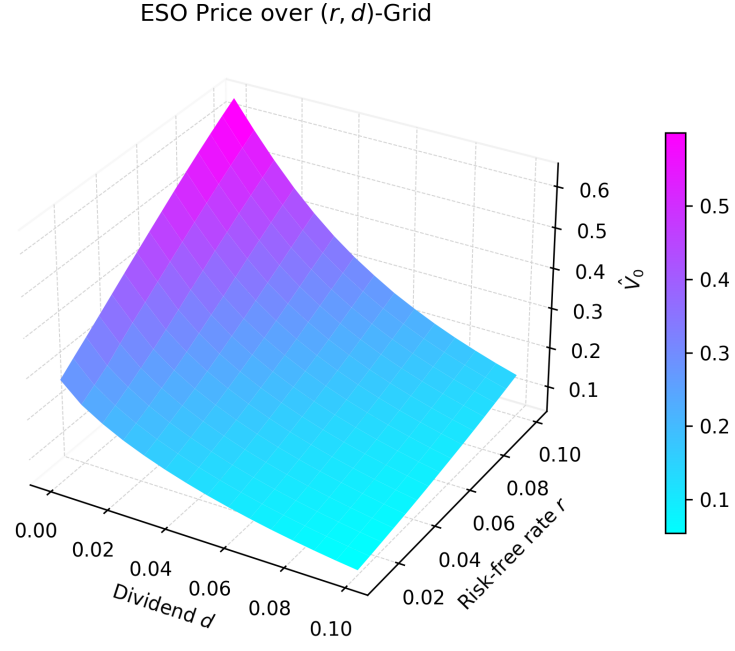


Figure 5: Estimated ESO price $\hat{V}_0$ over the (r, d) -grid.

5 Appendix

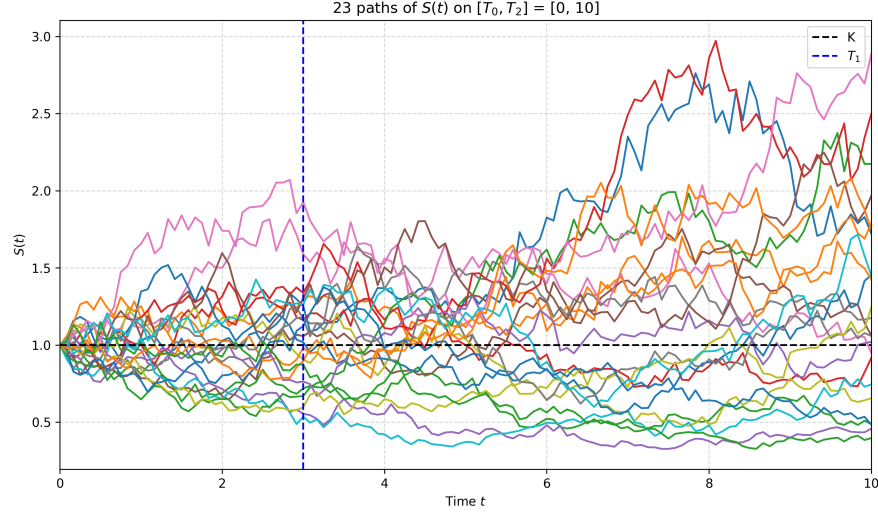


Figure 6: $N = 23$ simulated GBM paths of $S(t)$ on $[T_0, T_2] = [0, 10]$. The dashed vertical and horizontal lines mark the vesting boundary $T_1 = 3$ and strike $K = 1$, respectively.

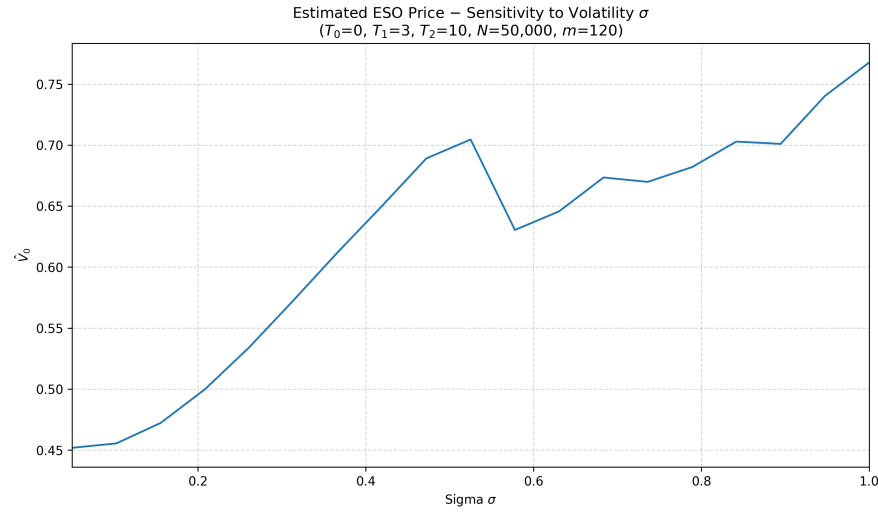


Figure 7: Estimated ESO price $\hat{V}_0$ as a function of volatility $\sigma \in [0.05, 1.0]$, with all other parameters held fixed.

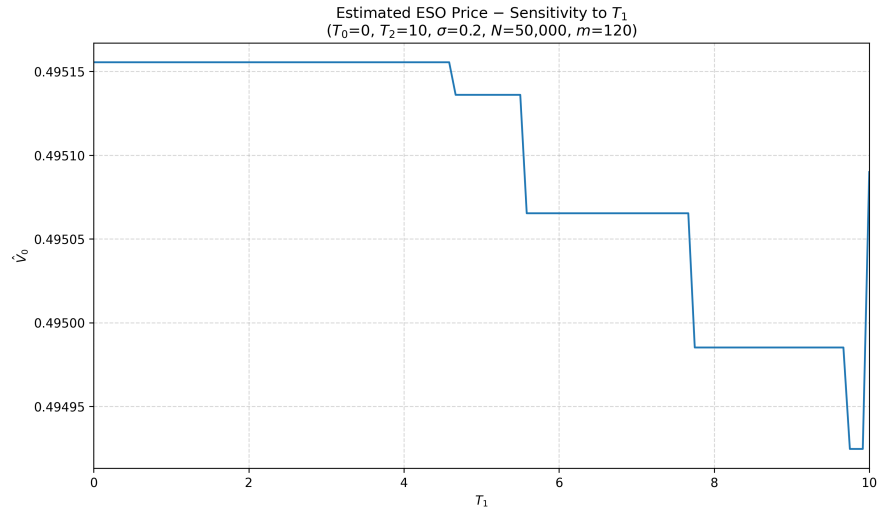


Figure 8: Estimated ESO price $\hat{V}_0$ as a function of $T_1 \in [0, T_2)$, with $T_2 = 10$ fixed.

References

- Oosterlee, C. W. and Grzelak, L. A. (2019). *Mathematical Modeling and Computation in Finance: With Exercises and Python and MATLAB Computer Codes*. World Scientific Publishing.
- Longstaff, F. A. and Schwartz, E. S. (2001). Valuing American options by simulation: A simple least-squares approach. *Review of Financial Studies*, 14(1), 113–147.
- Hull, J. C. (2022). *Options, Futures, and Other Derivatives* (11th ed.). Pearson.
- Carpenter, J. N. (1998). The exercise and valuation of executive stock options. *Journal of Financial Economics*, 48(2), 127–158.